Name
Date
Course

Precalculus Review

I) The Exponent Rules

II) The Three Most Important Types of Factoring

III) The Root Formulas

IV) Trigonometry

a) Radians
b) Trigonometric Ratios
c) Special Triangles
d) Graphs of Trigonometric Functions

The Exponent Rules

Exponent Rules Learned in Pre-Algebra

I) $x^a \cdot x^b = x^{a+b}$ Example: $x^2 \cdot x^5 = x^{2+5} = x^7$

II) $\frac{x^a}{x^b} = x^{a-b}$, $x \neq 0$ Example: $\frac{x^9}{x^3} = x^{9-3} = x^6$

New Exponent Rules

III) $(xy)^a = x^a \cdot y^a$ Example: $(2m)^4 = 2^4 \cdot m^4$

IV) $\left(\frac{x}{y}\right)^a = \frac{x^a}{y^a}$, $y \neq 0$ Example: $\left(\frac{n}{3}\right)^7 = \frac{n^7}{3^7}$

V) $(x^a)^b = x^{ab}$ Example: $(x^{10})^3 = x^{10 \cdot 3} = x^{30}$

Strange New Exponent Rules

VI) $x^1 = x$ Example: $3^1 = 3$

VII) $x^0 = 1$, $x \neq 0$ Example: $5^0 = 1$

VIII) $x^{-a} = \frac{1}{x^a}$, $x \neq 0$ Example: $x^{-2} = \frac{1}{x^2}$

IX) $x^{\frac{1}{a}} = \sqrt[a]{x}$ Example: $x^{\frac{1}{3}} = \sqrt[3]{x}$

X) $x^{\frac{b}{a}} = (\sqrt[a]{x})^b$ or $\sqrt[a]{x^b}$ Example: $x^{\frac{5}{7}} = (\sqrt[7]{x})^5$ or $\sqrt[7]{x^5}$

Simplify the following expressions. Do not leave your answers with negative exponents or fraction exponents.

① $\left(\frac{x^3y^7b^0y^5}{xa^4}\right)^2 b^{-\frac{2}{3}}$

I, VII, VI $\quad \left(\frac{x^3y^{12}(1)}{x^1a^4}\right)^2 b^{-\frac{2}{3}}$

II, VIII $\quad \left(\frac{x^2y^{12}}{a^4}\right)^2 \frac{1}{b^{2/3}}$

IV, X $\quad \frac{(x^2y^{12})^2}{(a^4)^2}\,\frac{1}{\sqrt[3]{b^2}}$

III $\quad \frac{(x^2)^2(y^{12})^2}{(a^4)^2\sqrt[3]{b^2}}$

V $\quad \boxed{\frac{x^4y^{24}}{a^8\sqrt[3]{b^2}}}$

② $\left(\frac{(4a)^0x^6y^2}{y^5b^6\cdot b}\right)^{-\frac{1}{2}}$

VII, II, VI $\quad \left(\frac{(1)y^{-3}x^6}{b^6\cdot b^1}\right)^{-\frac{1}{2}}$

I $\quad \left(\frac{y^{-3}x^6}{b^7}\right)^{-\frac{1}{2}}$

IV $\quad \frac{(y^{-3}x^6)^{-\frac{1}{2}}}{(b^7)^{-\frac{1}{2}}}$

III $\quad \frac{(y^{-3})^{-\frac{1}{2}}(x^6)^{-\frac{1}{2}}}{(b^7)^{-\frac{1}{2}}}$

V $\quad \frac{y^{\frac{3}{2}}x^{-3}}{b^{-\frac{7}{2}}}$

VIII, VIII', X $\quad \frac{b^{\frac{7}{2}}y^{\frac{3}{2}}}{x^3} = \boxed{\frac{\sqrt{b^7}\sqrt{y^3}}{x^3}}$

③ $\left(\frac{a}{b}\right)^{10}\sqrt[3]{\frac{b^6(2m)^0}{a^4b^3a^4}}$

IV, VII, II, I $\quad \left(\frac{a^{10}}{b^{10}}\right)\sqrt[3]{\frac{b^3(1)}{a^8}}$

IX $\quad \frac{a^{10}}{b^{10}}\left(\frac{b^3}{a^8}\right)^{\frac{1}{3}}$

IV $\quad \frac{a^{10}}{b^{10}}\,\frac{(b^3)^{\frac{1}{3}}}{(a^8)^{\frac{1}{3}}}$

V $\quad \frac{a^{10}}{b^{10}}\,\frac{b^1}{a^{8/3}}$

II $\quad b^{-9}a^{10-\frac{8}{3}}$

$b^{-9}a^{\frac{22}{3}}$

VIII, X $\quad \boxed{\frac{\sqrt[3]{a^{22}}}{b^9}}$

$4)\ \left(\frac{m^3 n^{10} m^{-2}}{n^6 L^{\frac{1}{16}}}\right)^4 \left(\frac{n^0}{m}\right)^{-1}$

I, II, VII

$$\left(\frac{m^1 n^4}{L^{\frac{1}{16}}}\right)^4 \left(\frac{1}{m}\right)^{-1}$$

IV

$$\frac{(m^1 n^4)^4}{(L^{\frac{1}{16}})^4} \quad \frac{1^{-1}}{m^{-1}}$$

III, VIII

$$\frac{(m^1)^4 (n^4)^4}{(L^{\frac{1}{16}})^4} \quad \frac{1}{m^{-1}\ 1^1}$$

V, VI

$$\frac{m^4 n^{16}}{L^{\frac{1}{4}}} \quad \frac{1}{m^{-1}(1)}$$

II, IX

$$\frac{m^4 n^{16}}{L^{1/4}\ m^{-1}}$$

$$\boxed{\frac{m^5 n^{16}}{\sqrt[4]{L}}}$$

The Distributive Rule

Multiply.

$5)\ 3x(x-2) = \boxed{3x^2-6x}$

$6)\ 4y^2(10y^6-9y+1) = \boxed{40y^8-36y^3+4y^2}$

$7)\ 8x^4(2x^4+3x^3) = \boxed{16x^8+24x^7}$

$8)\ 7b(-b^2+5b-7) = \boxed{-7b^3+35b^2-49b}$

Multiplying Binomials Using the FOIL Acronym

$9)\ (x+6)(x+4)$

$x^2+4x+6x+24$

$\boxed{x^2+10x+24}$

$10)\ (2y-1)(3y+11)$

$6y^2+22y-3y-11$

$\boxed{6y^2+19y-11}$

$11)\ (y-2)(y-2)$

$y^2-2y-2y+4$

$\boxed{y^2-4y+4}$

$12)\ (7x+3)(4x-9)$

$28x^2-63x+12x-27$

$\boxed{28x^2-51x-27}$

The Three Most Important Types of Factoring

I) GCF (Greatest Common Factor)

II) Quadratic Expressions (ax^2+bx+c)

III) DOTS (Difference of Two Squares)

Other Types
i) Grouping
ii) ax^4+bx^2+c
iii) Multivariate
iv) Sum/Difference of cubes

Factor the expressions below. Write the type of factoring used in the margin.

(13) $10x^5-8x^4+16x^3+6x^2$

GCF $\boxed{2x^2(5x^3-4x^2+8x+3)}$

(14) $-9x^8+45x-9x^2$

GCF $\boxed{-9x(x^7-5+x)}$

(15) x^2+2x-3

Quadratic $\boxed{(x-1)(x+3)}$

(16) $x^2-8x+15$

Quadratic $\boxed{(x-3)(x-5)}$

(17) $x^2+16x+55$

Quadratic $\boxed{(x+11)(x+5)}$

(18) x^2-6x-7

Quadratic $\boxed{(x+1)(x-7)}$

(19) $3x^2-2x-8$

Quadratic $\boxed{(3x+4)(x-2)}$

(20) $4x^2+25x+25$

Quadratic $\boxed{(4x+5)(x+5)}$

(21) $5x^2-12x+4$

Quadratic $\boxed{(5x-2)(x-2)}$

(22) $6x^2+11x-35$

Quadratic $\boxed{(2x+7)(3x-5)}$

(23) x^2-9

DOTS $\boxed{(x-3)(x+3)}$

(24) x^2-36

DOTS $\boxed{(x+6)(x-6)}$

DOTS (25) $64x^2-25=\boxed{(8x-5)(8x+5)}$

DOTS (26) $16x^2-1=\boxed{(4x-1)(4x+1)}$

The Root Formulas

Multiplication

$$\boxed{\sqrt[n]{a}\cdot\sqrt[n]{b}=\sqrt[n]{ab}}$$

Division

$$\boxed{\frac{\sqrt[n]{a}}{\sqrt[n]{b}}=\sqrt[n]{\frac{a}{b}}}$$

Simplify.

(27) $\sqrt{2}\cdot\sqrt{3}=\sqrt{2\cdot 3}=\boxed{\sqrt{6}}$

(28) $\sqrt{x}\cdot\sqrt{x}=\sqrt{x\cdot x}=\sqrt{x^2}=\boxed{x}$

(29) $\sqrt[3]{\frac{8}{27}}=\frac{\sqrt[3]{8}}{\sqrt[3]{27}}=\boxed{\frac{2}{3}}$

(30) $\sqrt{5}\cdot\sqrt{6}=\sqrt{5\cdot 6}=\boxed{\sqrt{30}}$

(31) $\sqrt{7}\cdot\sqrt{7}=\sqrt{7\cdot 7}=\sqrt{49}=\boxed{7}$

(32) $\sqrt{x+h}\cdot\sqrt{x+h}=\sqrt{(x+h)(x+h)}=\sqrt{(x+h)^2}=\boxed{x+h}$

(33) $\sqrt{\frac{16}{25}}=\frac{\sqrt{16}}{\sqrt{25}}=\boxed{\frac{4}{5}}$

(34) $\sqrt{18}=\sqrt{9\cdot 2}=\sqrt{9}\cdot\sqrt{2}=3\cdot\sqrt{2}=\boxed{3\sqrt{2}}$

(35) $\sqrt[3]{24}=\sqrt[3]{8\cdot 3}=\sqrt[3]{8}\cdot\sqrt[3]{3}=\boxed{2\sqrt[3]{3}}$

(36) $\sqrt{28} = \sqrt{4\cdot 7} = \sqrt{4}\cdot\sqrt{7} = 2\cdot\sqrt{7} = \boxed{2\sqrt{7}}$

(37) $\sqrt[4]{64} = \sqrt[4]{16\cdot 4} = \sqrt[4]{16}\cdot\sqrt[4]{4} = \boxed{2\sqrt[4]{4}}$

(38) $\frac{7}{\sqrt{6}} = \frac{7\cdot\sqrt{6}}{\sqrt{6}\cdot\sqrt{6}} = \frac{7\sqrt{6}}{\sqrt{6\cdot 6}} = \frac{7\sqrt{6}}{\sqrt{36}} = \boxed{\frac{7\sqrt{6}}{6}}$

(39) $-\frac{2}{\sqrt{12}} = -\frac{2}{\sqrt{4\cdot 3}} = -\frac{2}{\sqrt{4}\cdot\sqrt{3}} = -\frac{\cancel{2}}{\cancel{2}\cdot\sqrt{3}} = -\frac{1}{\sqrt{3}} = -\frac{1\cdot\sqrt{3}}{\sqrt{3}\cdot\sqrt{3}}$

$= -\frac{\sqrt{3}}{\sqrt{3\cdot 3}}$

$= -\frac{\sqrt{3}}{\sqrt{9}}$

$= \boxed{-\frac{\sqrt{3}}{3}}$

(40) $-\frac{3}{\sqrt{2}} = -\frac{3\cdot\sqrt{2}}{\sqrt{2}\cdot\sqrt{2}} = -\frac{3\sqrt{2}}{\sqrt{2\cdot 2}} = -\frac{3\sqrt{2}}{\sqrt{4}} = \boxed{-\frac{3\sqrt{2}}{2}}$

(41) $\frac{10}{\sqrt{50}} = \frac{10}{\sqrt{25\cdot 2}} = \frac{10}{\sqrt{25}\cdot\sqrt{2}} = \frac{10}{5\cdot\sqrt{2}} = \frac{2}{\sqrt{2}} = \frac{2\cdot\sqrt{2}}{\sqrt{2}\cdot\sqrt{2}} = \frac{2\sqrt{2}}{\sqrt{2\cdot 2}}$

$= \frac{2\sqrt{2}}{\sqrt{4}}$

$= \frac{2\sqrt{2}}{2}$

$= \boxed{\sqrt{2}}$

(42) $\sqrt{\frac{18}{5}} = \frac{\sqrt{18}}{\sqrt{5}} = \frac{\sqrt{9\cdot 2}}{\sqrt{5}} = \frac{\sqrt{9}\sqrt{2}}{\sqrt{5}} = \frac{3\sqrt{2}\cdot\sqrt{5}}{\sqrt{5}\cdot\sqrt{5}} = \frac{3\sqrt{2\cdot 5}}{\sqrt{5\cdot 5}} = \frac{3\sqrt{10}}{\sqrt{25}}$

$= \boxed{\frac{3\sqrt{10}}{5}}$

(43) $\sqrt{\frac{32}{11}} = \frac{\sqrt{32}}{\sqrt{11}} = \frac{\sqrt{16\cdot 2}}{\sqrt{11}} = \frac{\sqrt{16}\cdot\sqrt{2}}{\sqrt{11}} = \frac{4\cdot\sqrt{2}}{\sqrt{11}} = \frac{4\sqrt{2}\cdot\sqrt{11}}{\sqrt{11}\cdot\sqrt{11}}$

$= \frac{4\sqrt{2\cdot 11}}{\sqrt{11\cdot 11}}$

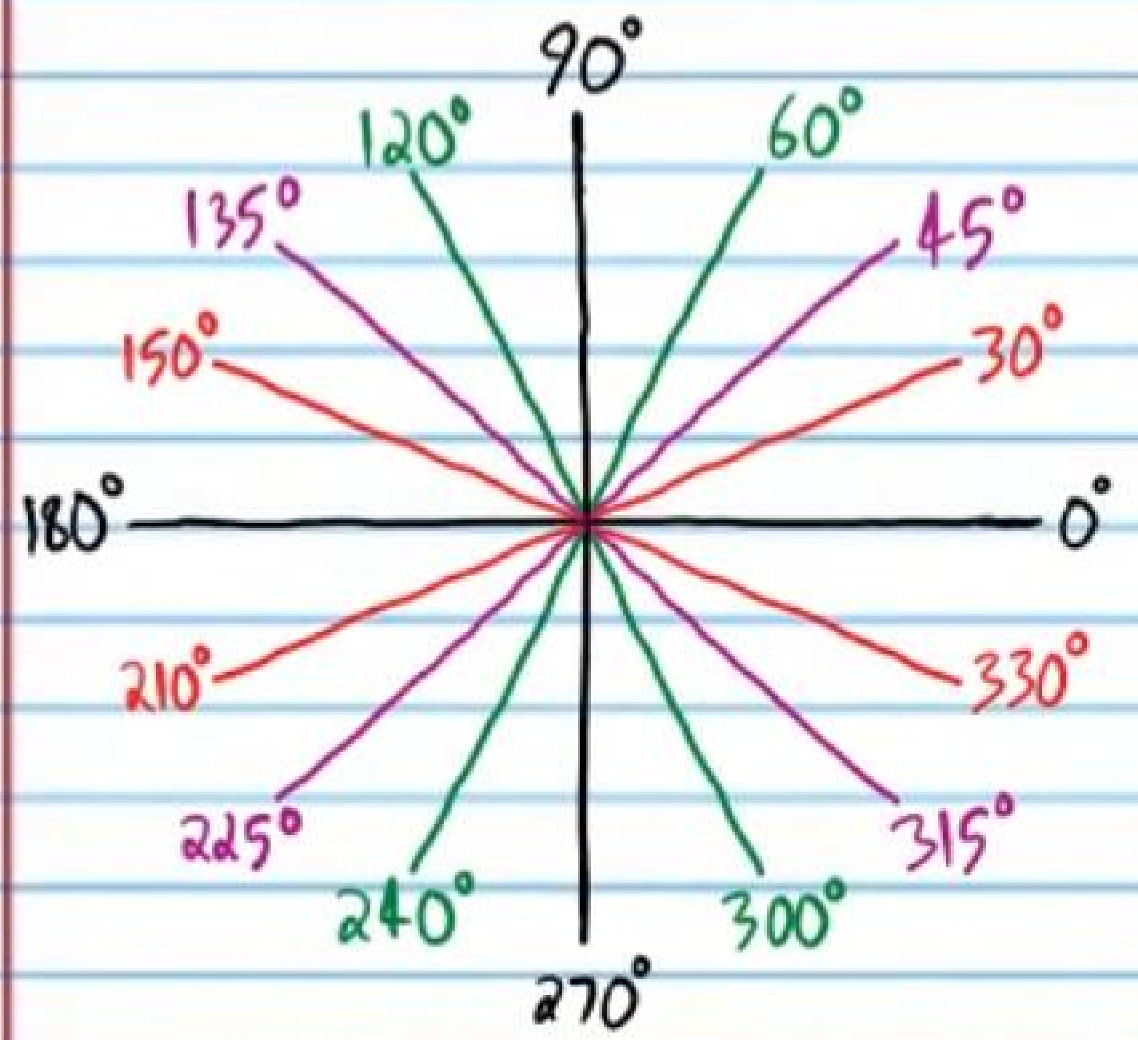
$= \frac{4\sqrt{22}}{\sqrt{121}}$

$= \boxed{\frac{4\sqrt{22}}{11}}$

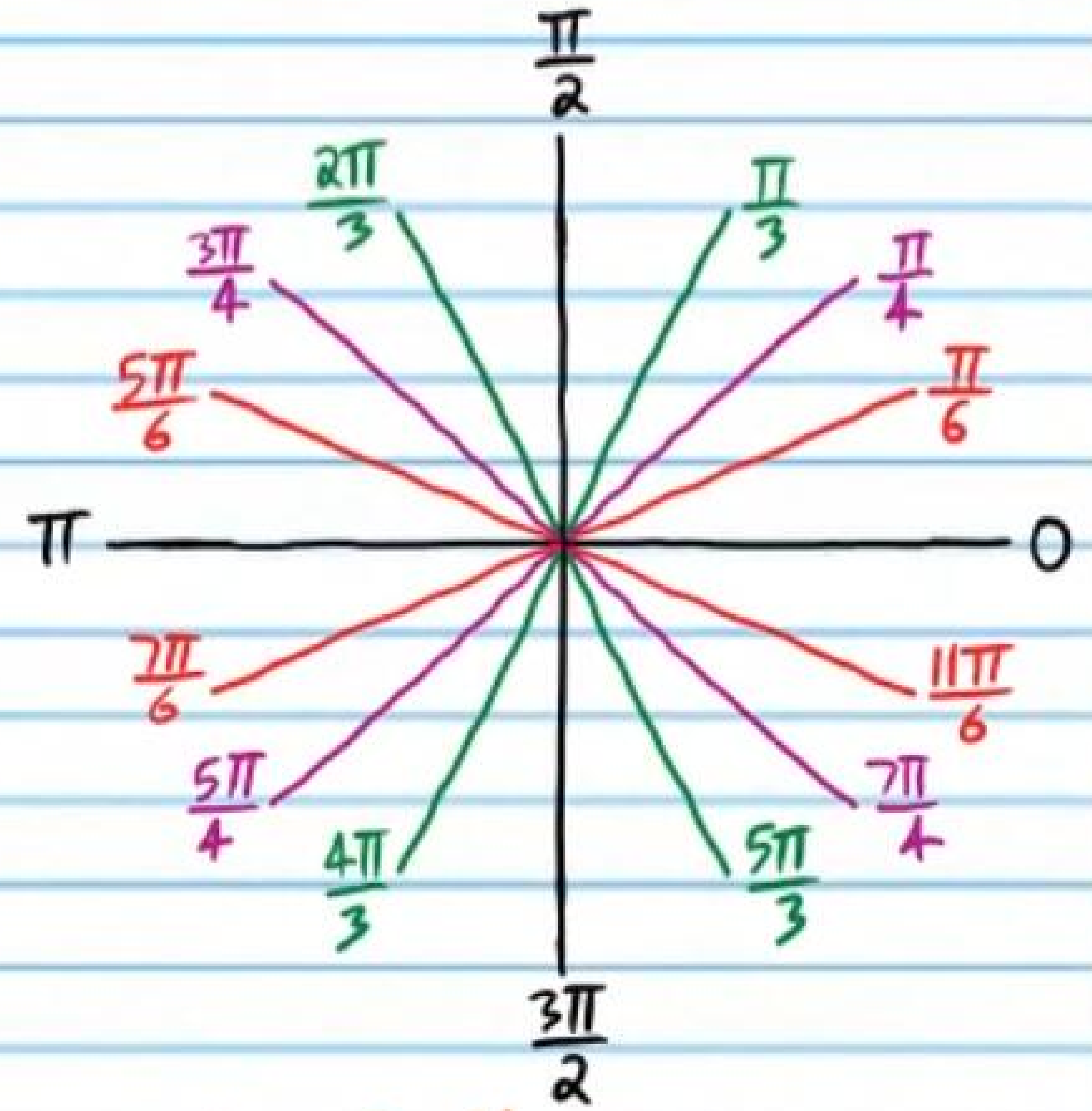
Trigonometry

In the study of calculus, we use radian units for the measures of angles (with a few exceptions).

Degree Units



Radian Units



Converting from Degrees to Radians

$$\text{degree measure} \times \frac{\pi \text{ radians}}{180 \text{ degrees}} = \text{radian measure}$$

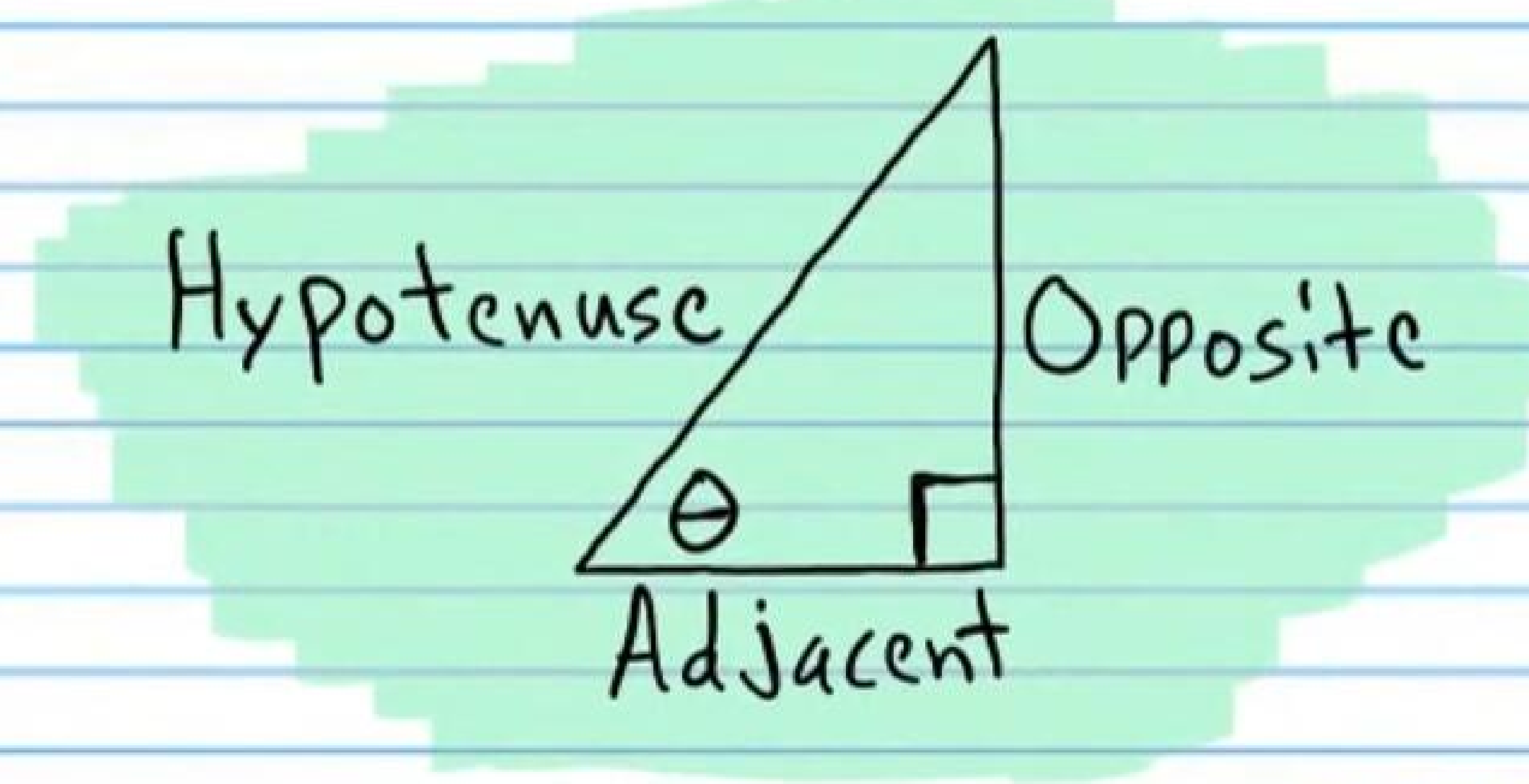
Example: $36^\circ \times \frac{\pi}{180^\circ} = \frac{36\pi}{180} = \boxed{\frac{\pi}{5}}$

Converting from Radians to Degrees

$$\text{Radian measure} \times \frac{180 \text{ degrees}}{\pi \text{ radians}} = \text{degree measure}$$

Example: $\frac{2\pi}{3} \times \frac{180^\circ}{\pi} = \frac{360\pi^\circ}{3\pi} = \frac{360^\circ}{3} = \boxed{120^\circ}$

The Six Basic Trigonometric Functions



$\sin\theta = \frac{\text{Opp}}{\text{Hyp}}$ | $\csc\theta = \frac{\text{Hyp}}{\text{Opp}}$

$\cos\theta = \frac{\text{Adj}}{\text{Hyp}}$ | $\sec\theta = \frac{\text{Hyp}}{\text{Adj}}$

$\tan\theta = \frac{\text{Opp}}{\text{Adj}}$ | $\cot\theta = \frac{\text{Adj}}{\text{Opp}}$

SOHCAHTOA

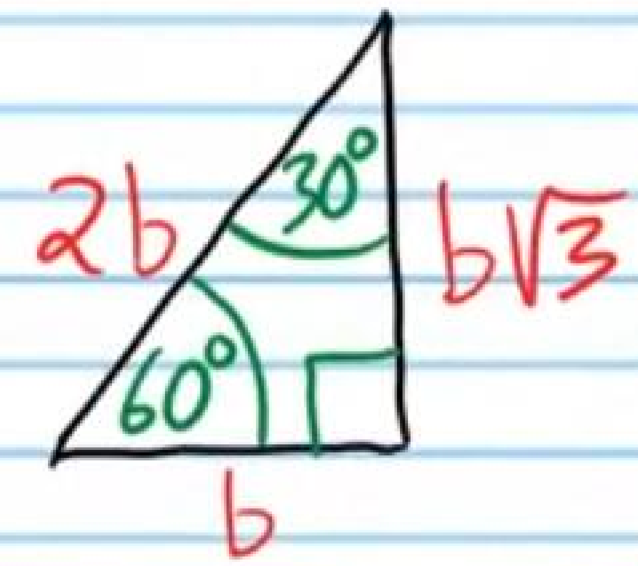
S – sine, O – opposite, H – Hypotenuse

C – Cosine, A – Adjacent, H – Hypotenuse

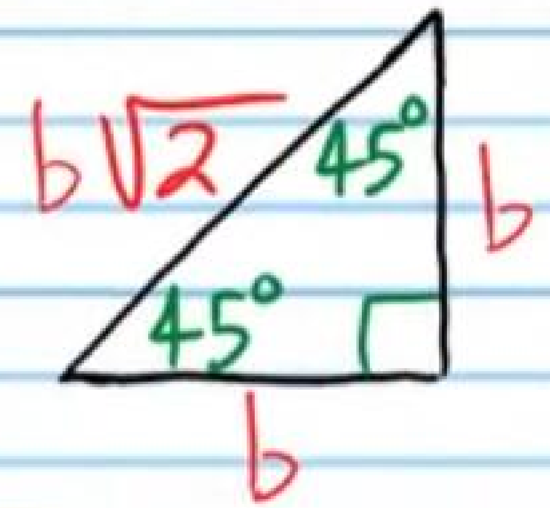
T – tangent, O – opposite, A – Adjacent

Special Right Triangles

30°-60°-90° Triangle Theorem: If the inner angles of a triangle are 30°, 60°, and 90°, and the length of the side opposite the 30° angle is b, then the length of the hypotenuse is $2b$ and the length of the side opposite the 60° angle is $b\sqrt{3}$.

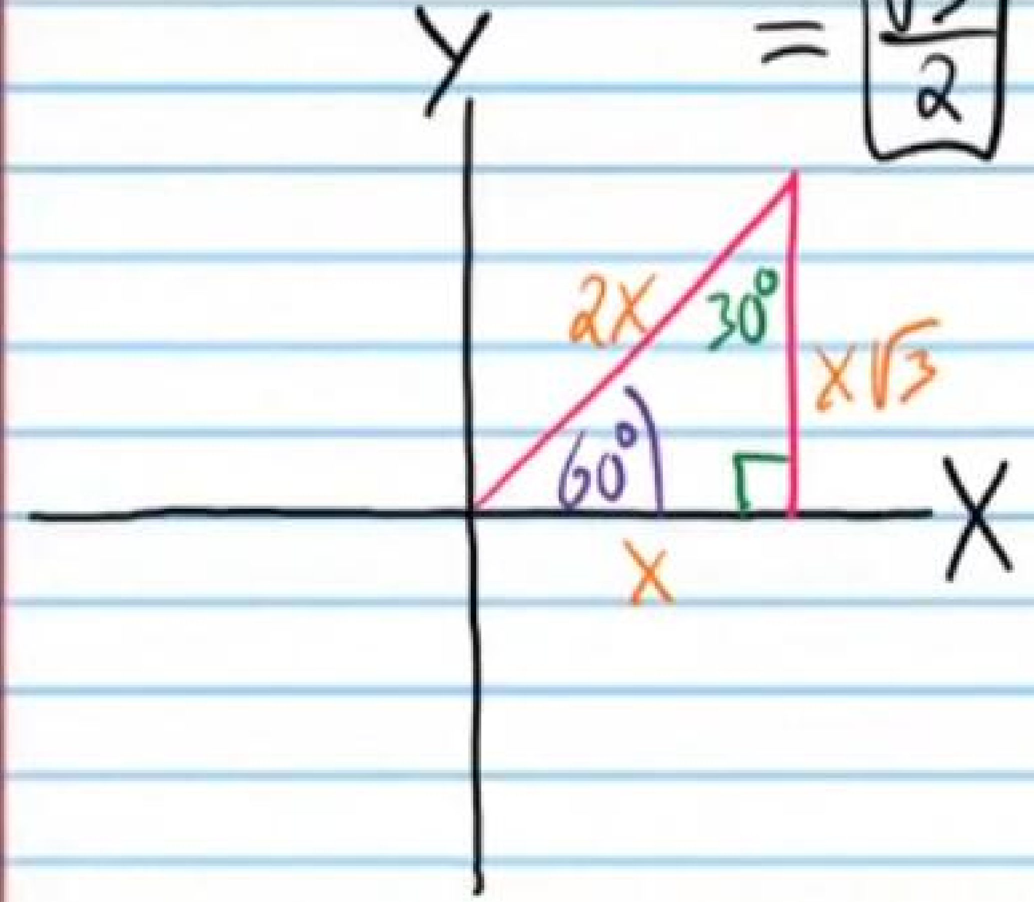


45°-45°-90° Triangle Theorem: If the inner angles of a triangle are 45°, 45°, and 90°, and the length of the two sides opposite the 45° angles are b, then the length of the hypotenuse is $b\sqrt{2}$.

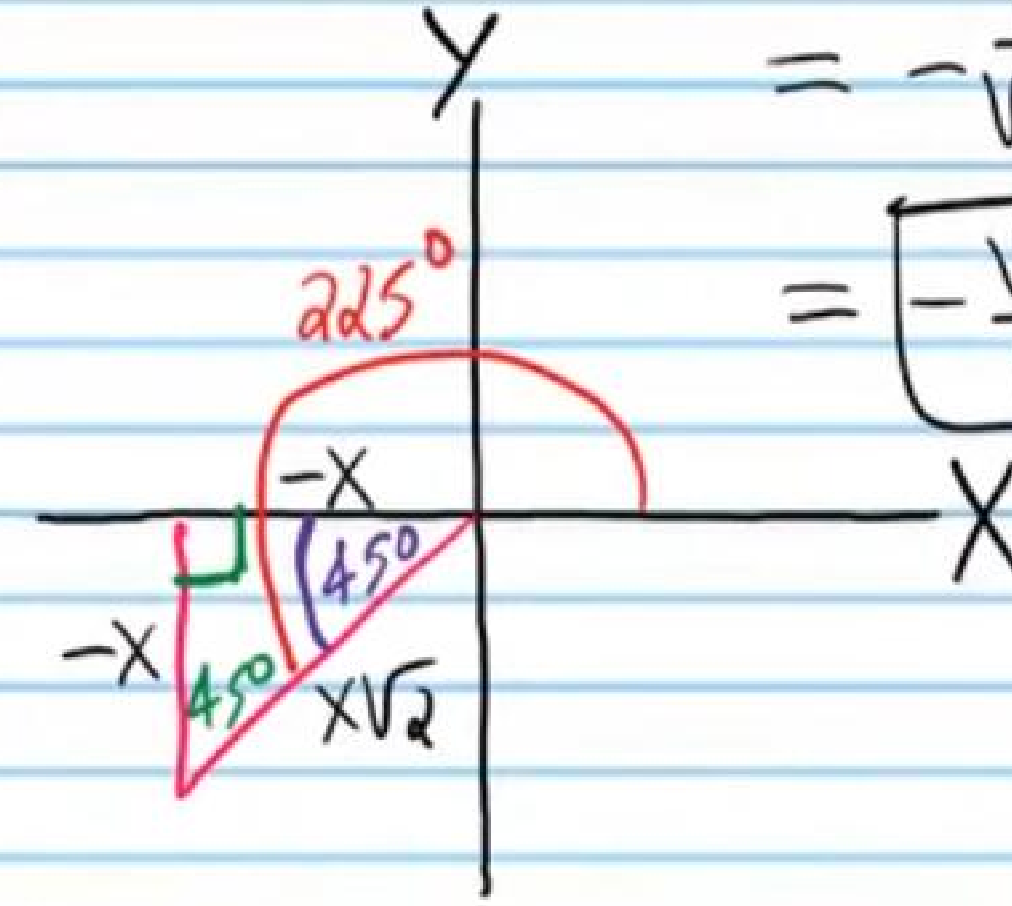


Find the value of each expression below.

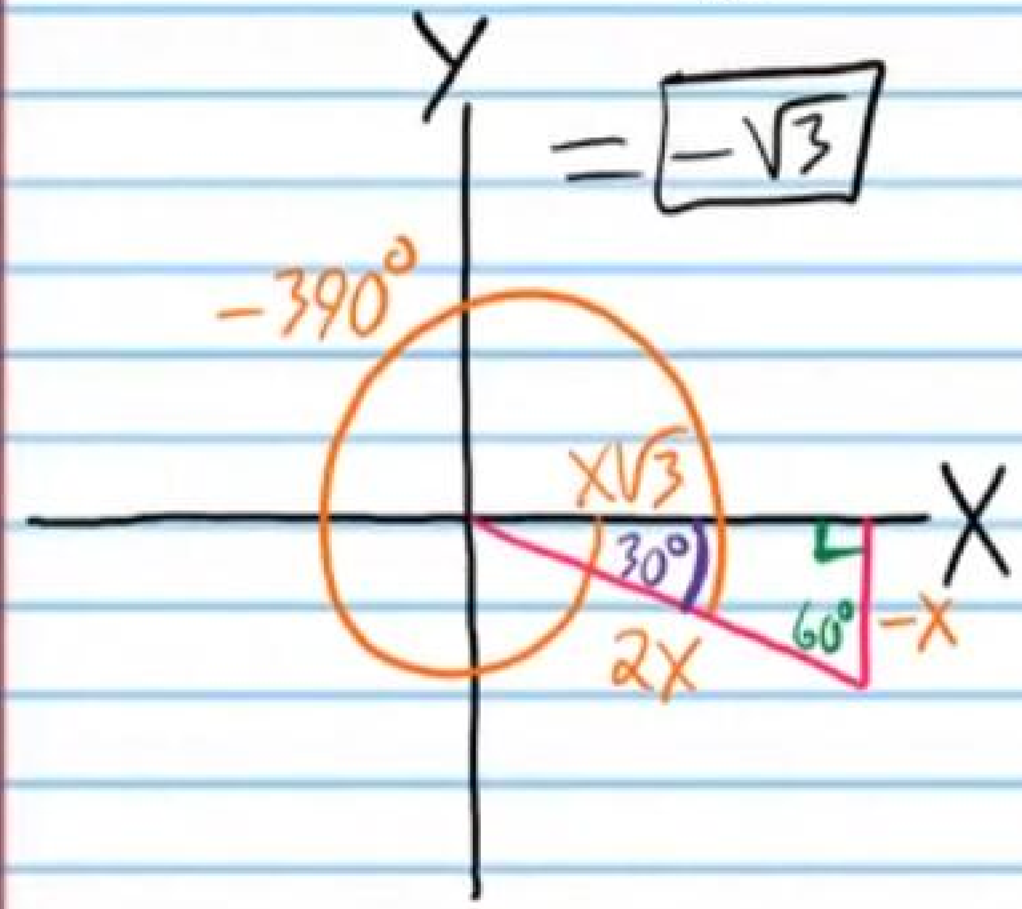
(44) $\sin\frac{\pi}{3} = \sin 60° = \frac{x\sqrt{3}}{2x} = \boxed{\frac{\sqrt{3}}{2}}$



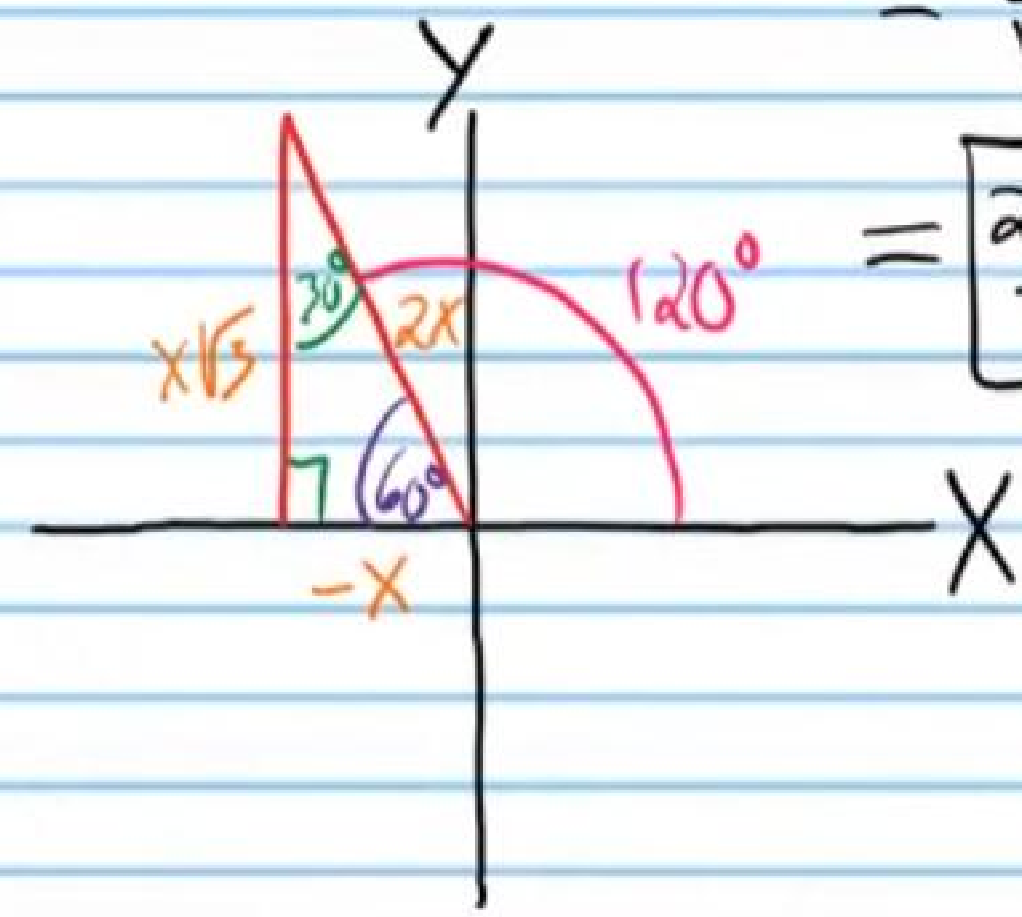
(45) $\cos\left(\frac{5\pi}{4}\right) = \cos 225° = \frac{-x}{x\sqrt{2}} = -\frac{1}{\sqrt{2}} = \boxed{-\frac{\sqrt{2}}{2}}$



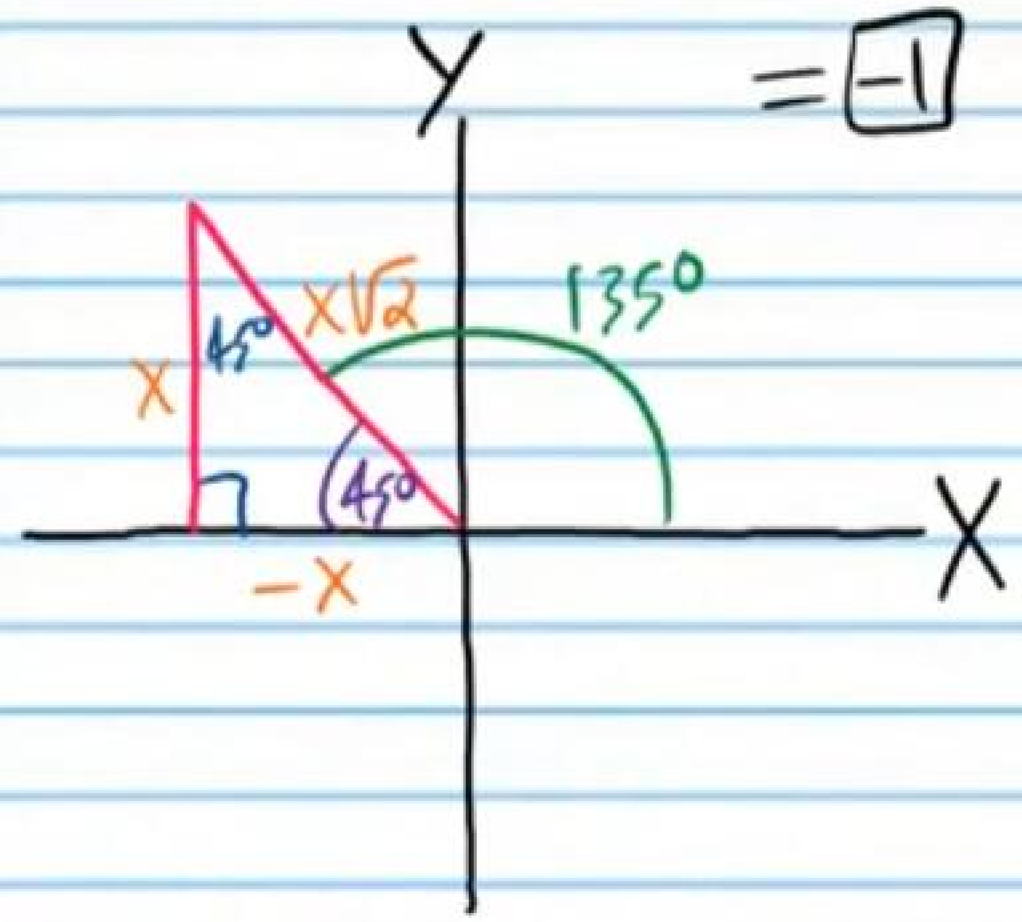
(46) $\cot\left(-\frac{13\pi}{6}\right) = \cot(-390°) = \frac{x\sqrt{3}}{-x} = \boxed{-\sqrt{3}}$



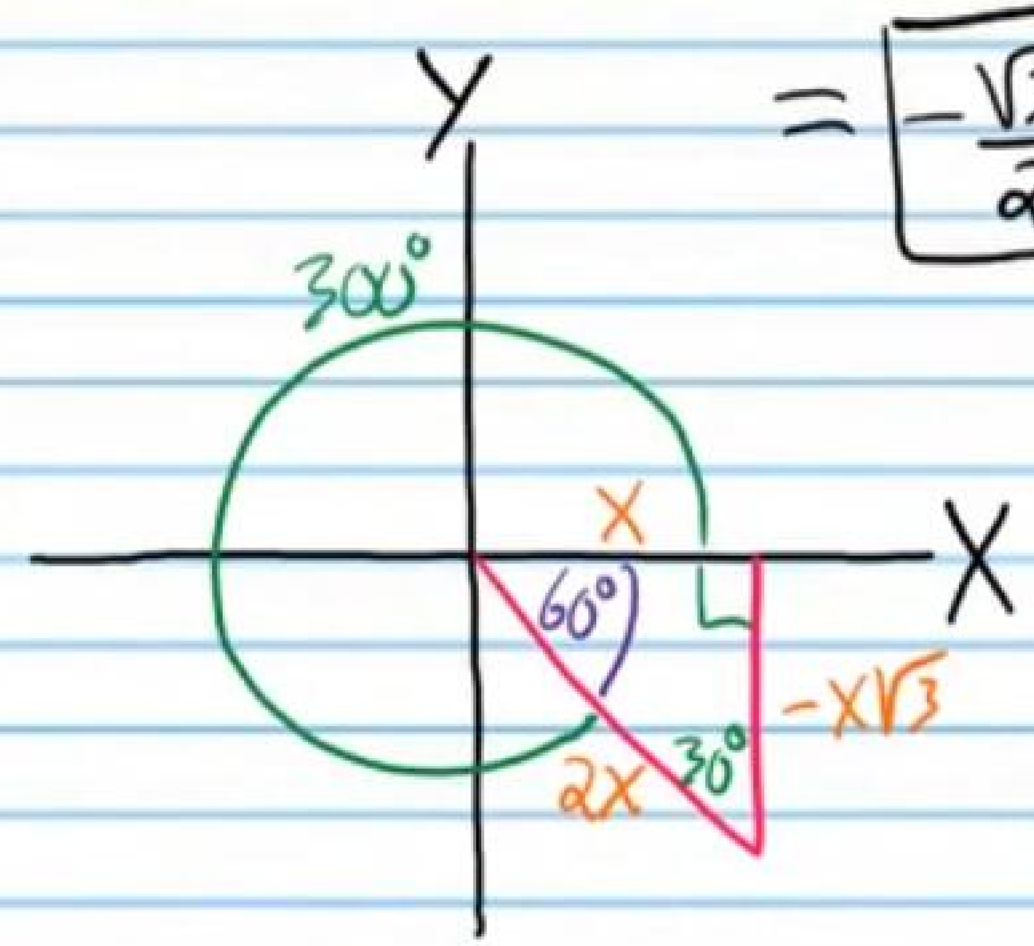
(47) $\csc\left(\frac{2\pi}{3}\right) = \csc 120° = \frac{2x}{x\sqrt{3}} = \frac{2}{\sqrt{3}} = \boxed{\frac{2\sqrt{3}}{3}}$



(48) $\tan\left(\frac{3\pi}{4}\right) = \tan 135^\circ = \frac{x}{-x} = \boxed{-1}$



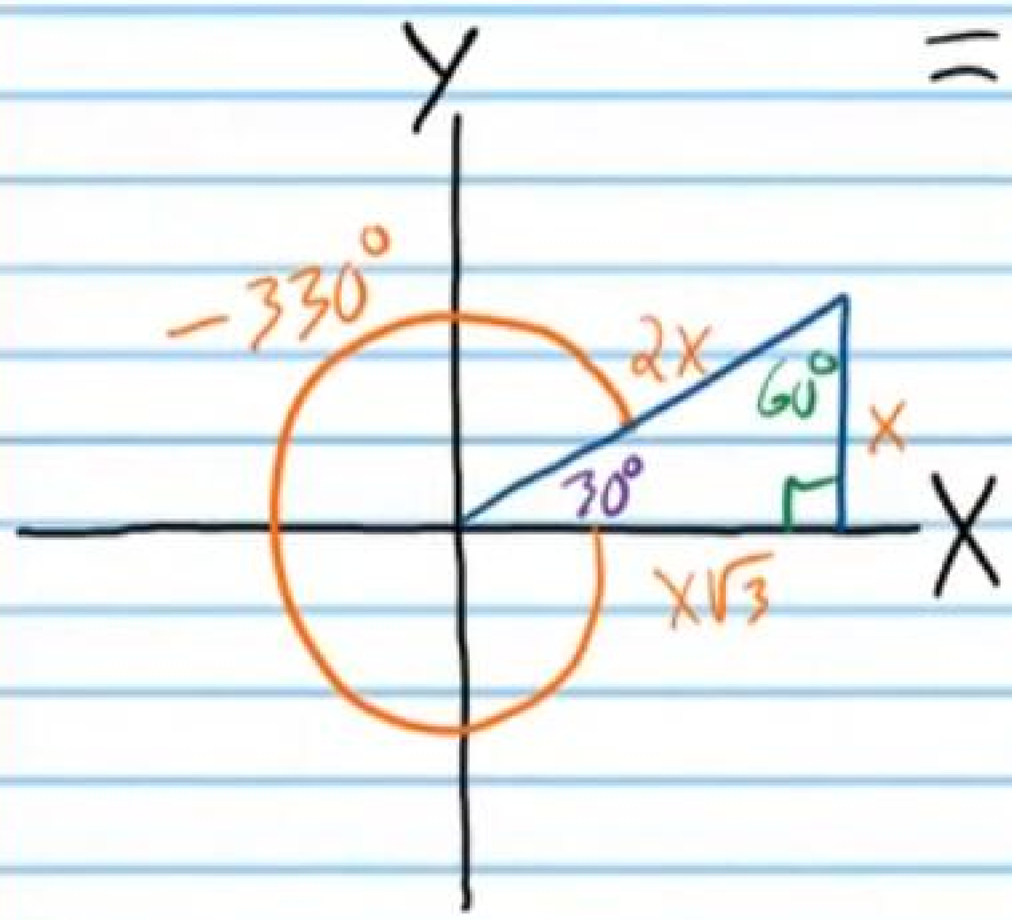
(49) $\sin\left(\frac{5\pi}{3}\right) = \sin 300^\circ = \frac{-x\sqrt{3}}{2x} = \boxed{-\frac{\sqrt{3}}{2}}$



(50) $\sec\left(-\frac{11\pi}{6}\right) = \sec(-330^\circ) = \frac{2x}{x\sqrt{3}}$

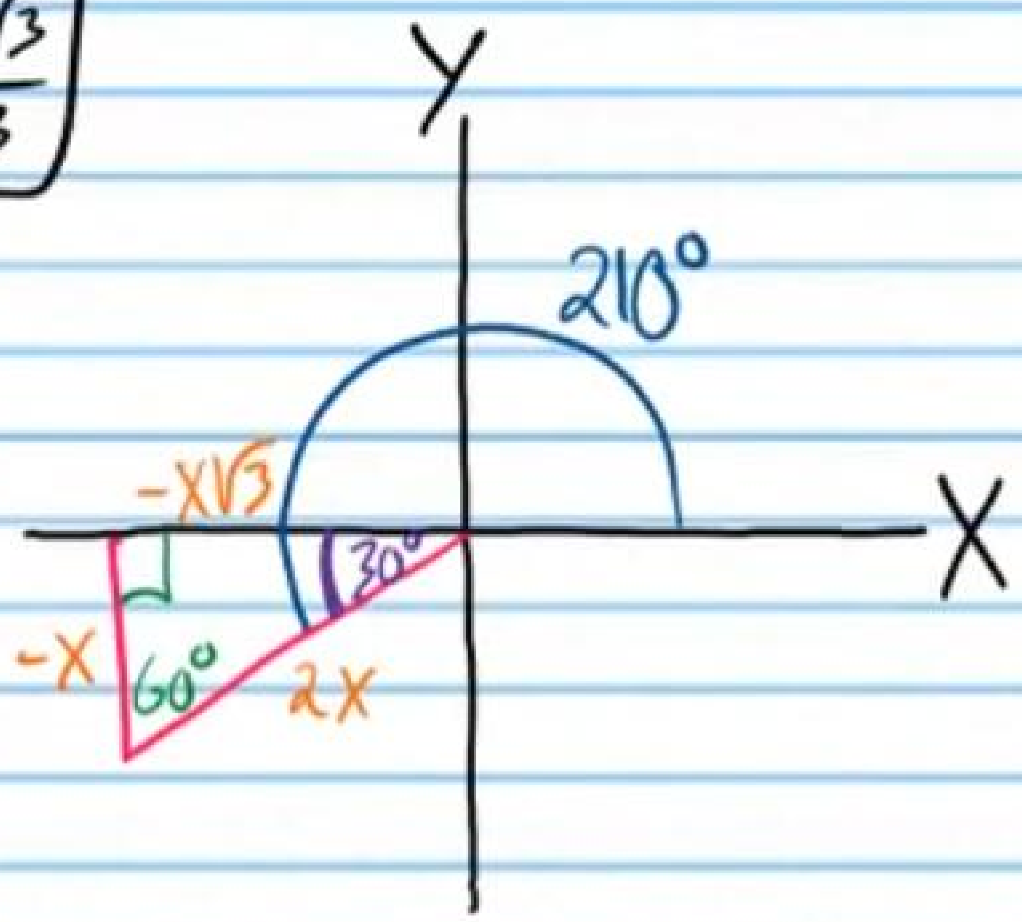
$= \frac{2}{\sqrt{3}}$

$= \boxed{\frac{2\sqrt{3}}{3}}$



(51) $\csc\left(\frac{7\pi}{6}\right) = \csc(210^\circ) = \frac{2x}{-x}$

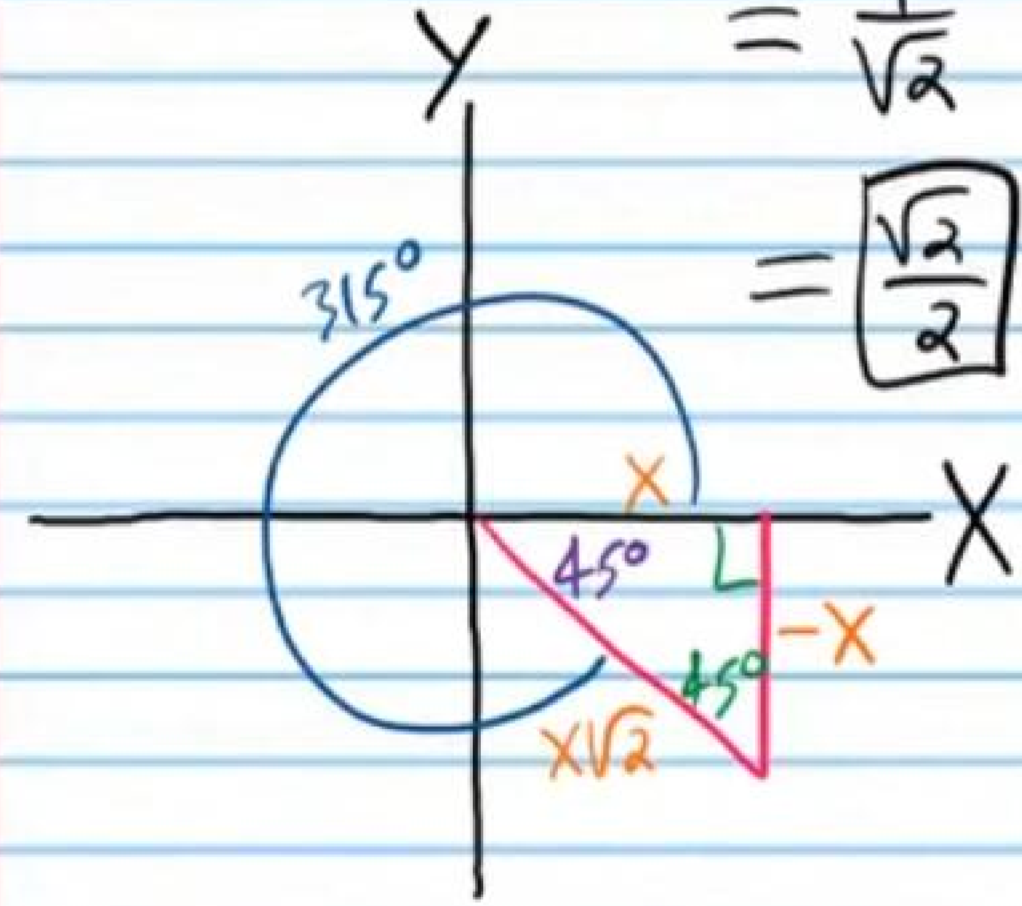
$= \boxed{-2}$



(52) $\cos\left(\frac{7\pi}{4}\right) = \cos 315^\circ = \frac{x}{x\sqrt{2}}$

$= \frac{1}{\sqrt{2}}$

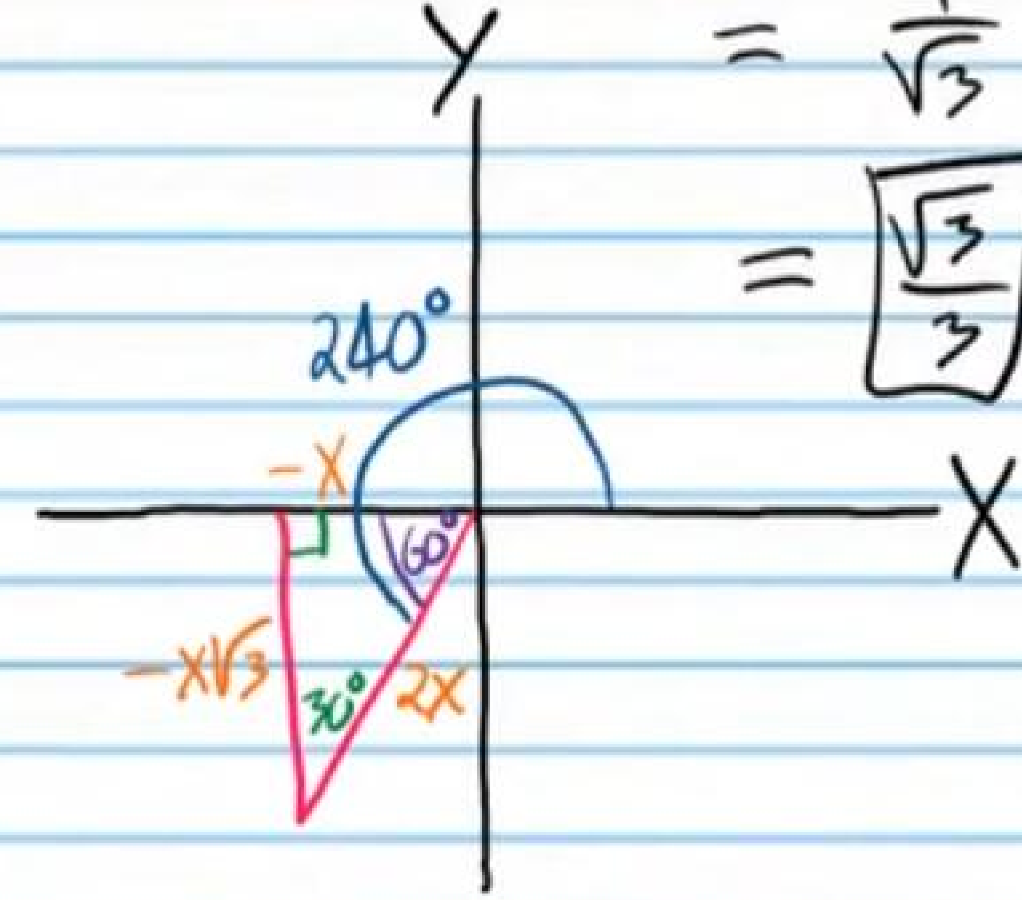
$= \boxed{\frac{\sqrt{2}}{2}}$



(53) $\cot\left(\frac{4\pi}{3}\right) = \cot 240^\circ = \frac{-x}{-x\sqrt{3}}$

$= \frac{1}{\sqrt{3}}$

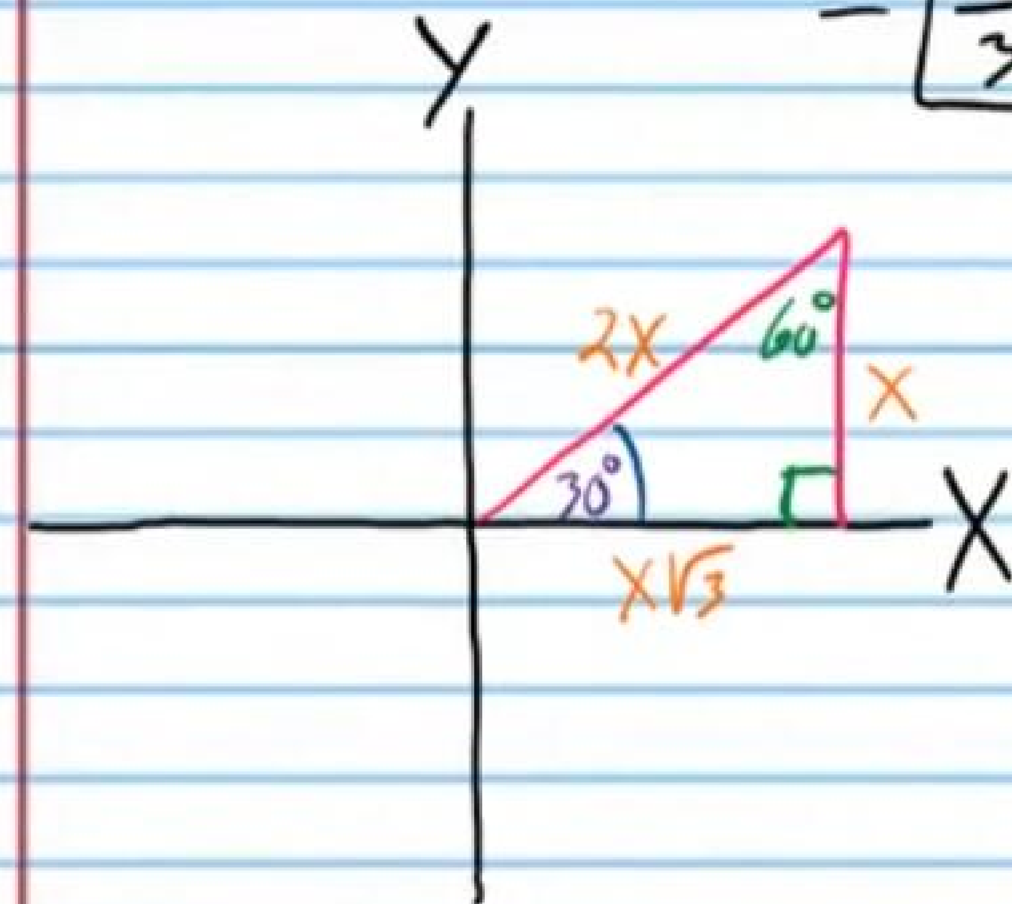
$= \boxed{\frac{\sqrt{3}}{3}}$



(54) $\tan\left(\frac{\pi}{6}\right) = \tan 30^\circ = \frac{x}{x\sqrt{3}}$

$= \frac{1}{\sqrt{3}}$

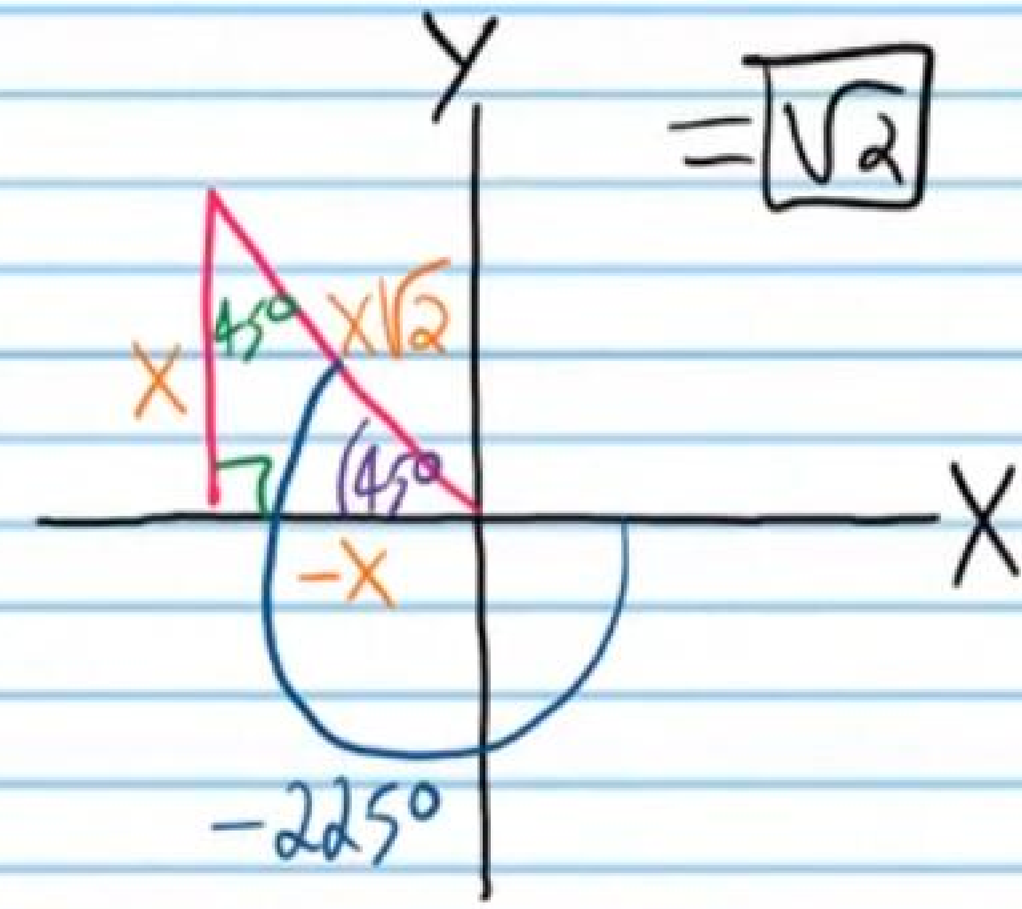
$= \boxed{\frac{\sqrt{3}}{3}}$



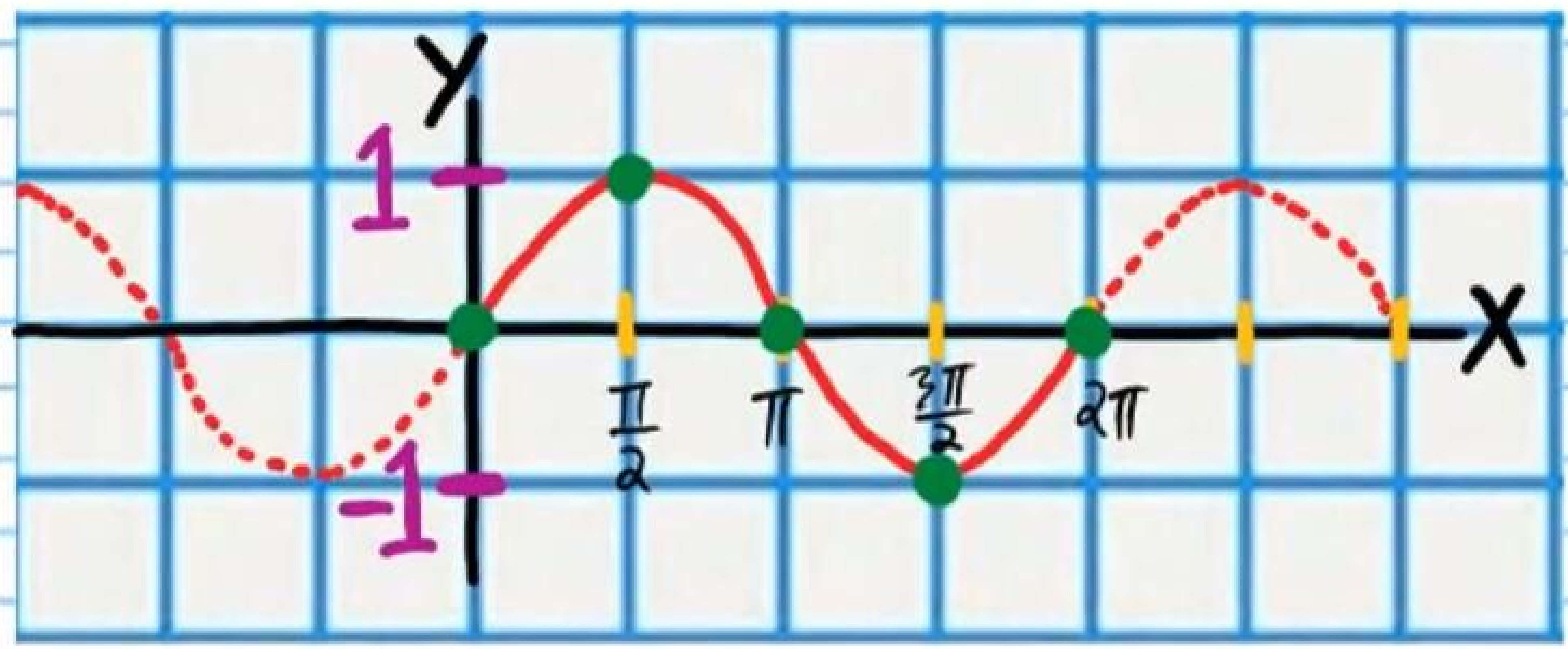
(55) $\csc\left(-\frac{5\pi}{4}\right) = \csc(-225^\circ)$

$= \frac{x\sqrt{2}}{x}$

$= \boxed{\sqrt{2}}$



$y = \sin x$

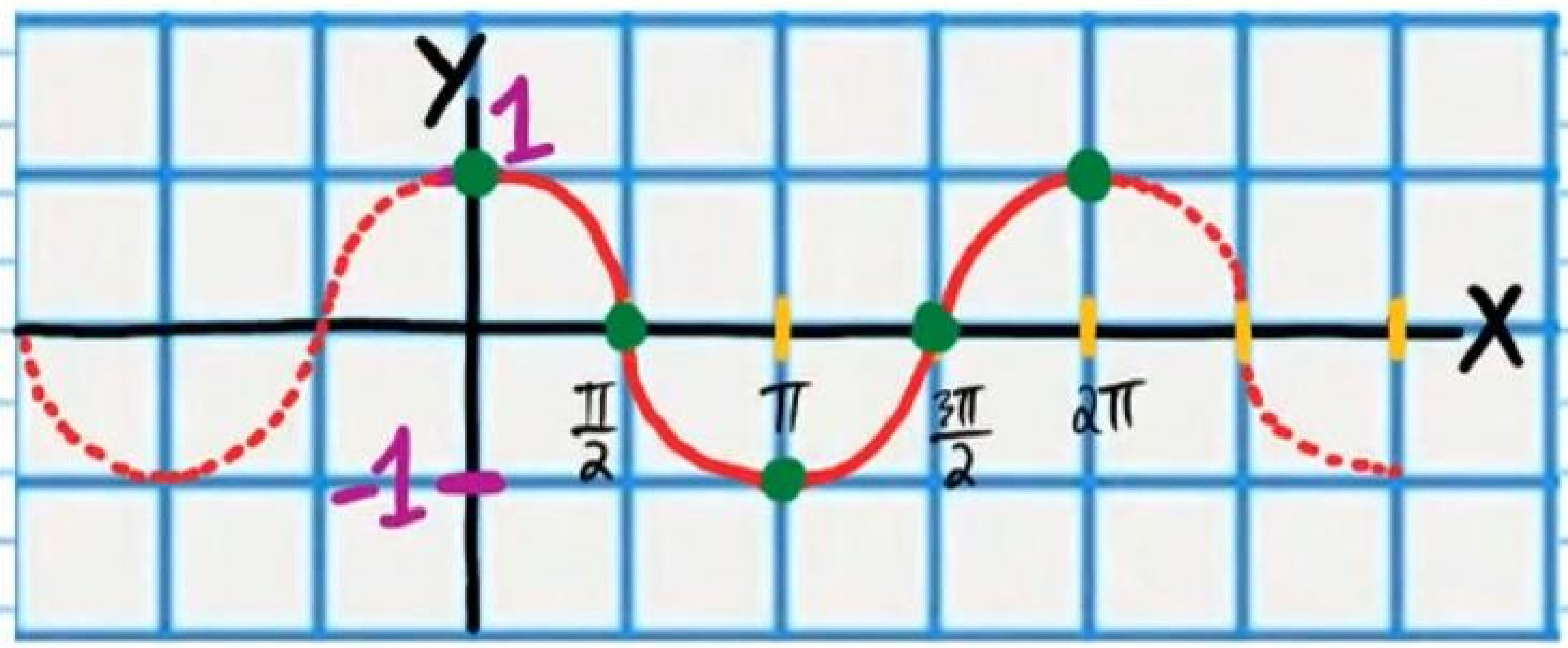


Use the graph above to find the following values.

56) $\sin\left(\frac{3\pi}{2}\right) = \boxed{-1}$ 57) $\sin(2\pi) = \boxed{0}$ 58) $\sin(0) = \boxed{0}$

59) $\sin\left(-\frac{\pi}{2}\right) = \boxed{-1}$ 60) $\sin(-\pi) = \boxed{0}$ 61) $\sin\left(\frac{5\pi}{2}\right) = \boxed{1}$

$y = \cos x$

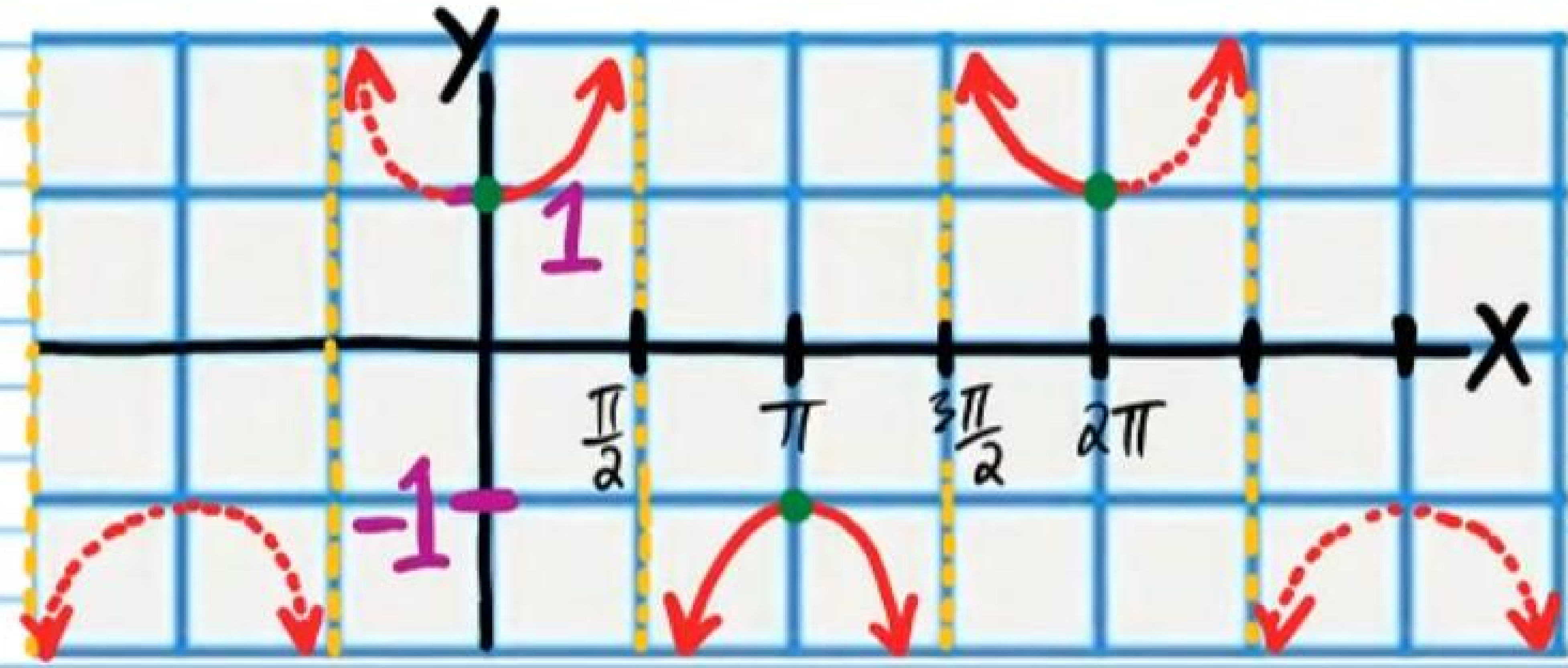


Use the graph above to find the following values.

62) $\cos\left(\frac{\pi}{2}\right) = \boxed{0}$ 63) $\cos \pi = \boxed{-1}$ 64) $\cos\left(\frac{3\pi}{2}\right) = \boxed{0}$

65) $\cos(2\pi) = \boxed{1}$ 66) $\cos 0 = \boxed{1}$ 67) $\cos\left(-\frac{\pi}{2}\right) = \boxed{0}$

$y = \sec x$

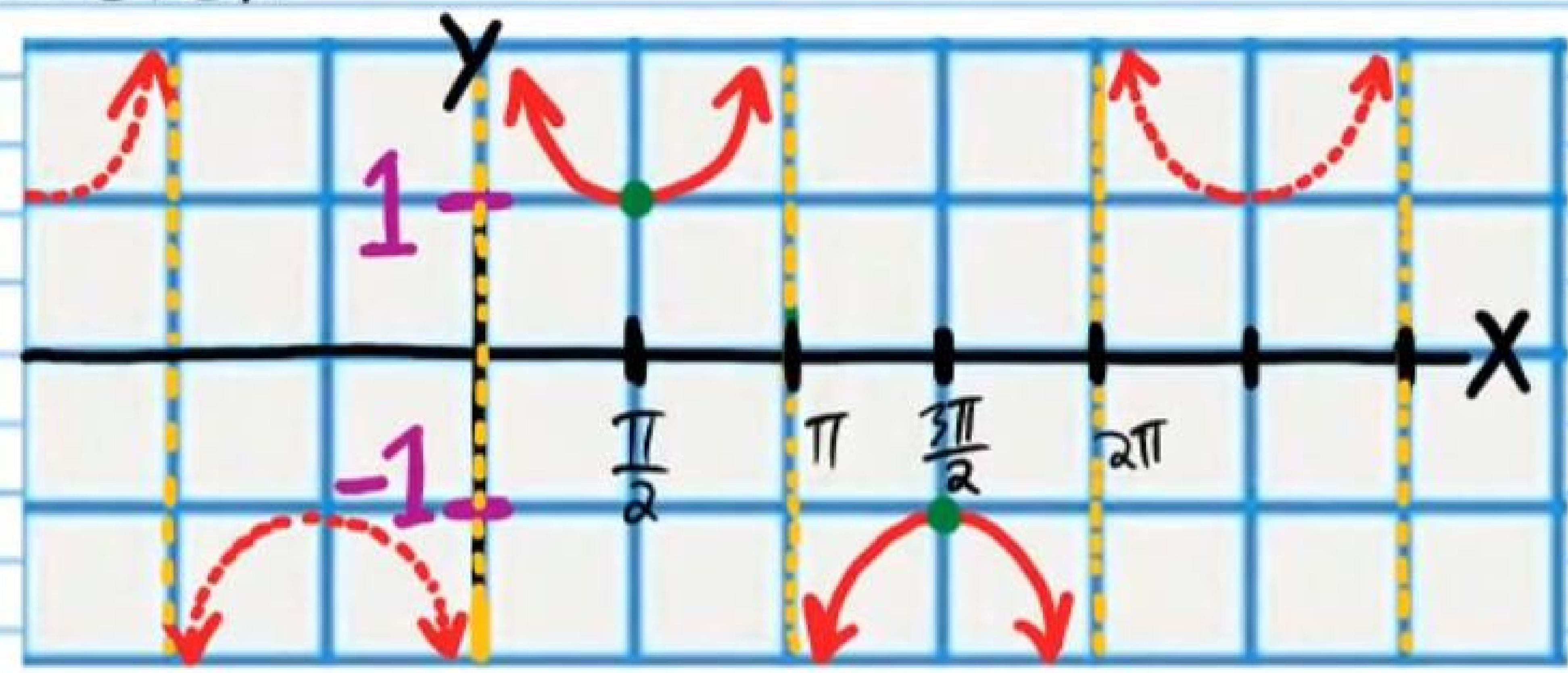


Use the graph above to find the following values.

68) $\sec \pi = \boxed{-1}$ 69) $\sec \frac{\pi}{2} = \boxed{\text{und}}$ 70) $\sec\left(-\frac{\pi}{2}\right) = \boxed{\text{und}}$

71) $\sec 0 = \boxed{1}$ 72) $\sec\left(\frac{3\pi}{2}\right) = \boxed{\text{und}}$ 73) $\sec(-\pi) = \boxed{-1}$

$y = \csc x$

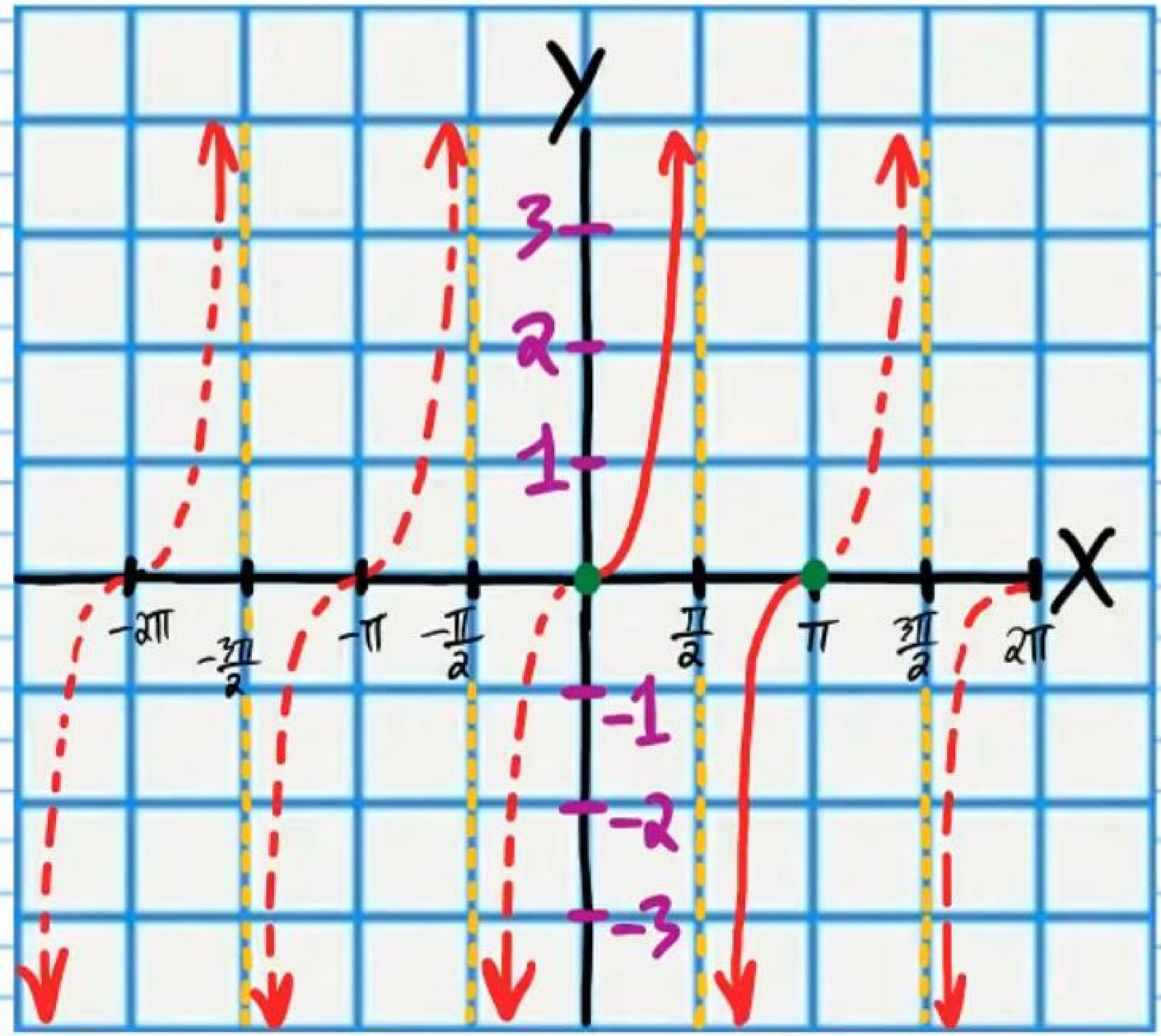


Use the graph above to find the following values.

74) $\csc\left(\frac{\pi}{2}\right) = \boxed{1}$ 75) $\csc \pi = \boxed{\text{und}}$ 76) $\csc\left(\frac{3\pi}{2}\right) = \boxed{-1}$

77) $\csc 0 = \boxed{\text{und}}$ 78) $\csc\left(-\frac{\pi}{2}\right) = \boxed{-1}$ 79) $\csc(2\pi) = \boxed{\text{und}}$

$y = \tan x$



Use the graph above to find the following values.

(80) $\tan \pi = \boxed{0}$

(81) $\tan\left(\frac{\pi}{2}\right) = \boxed{\text{und}}$

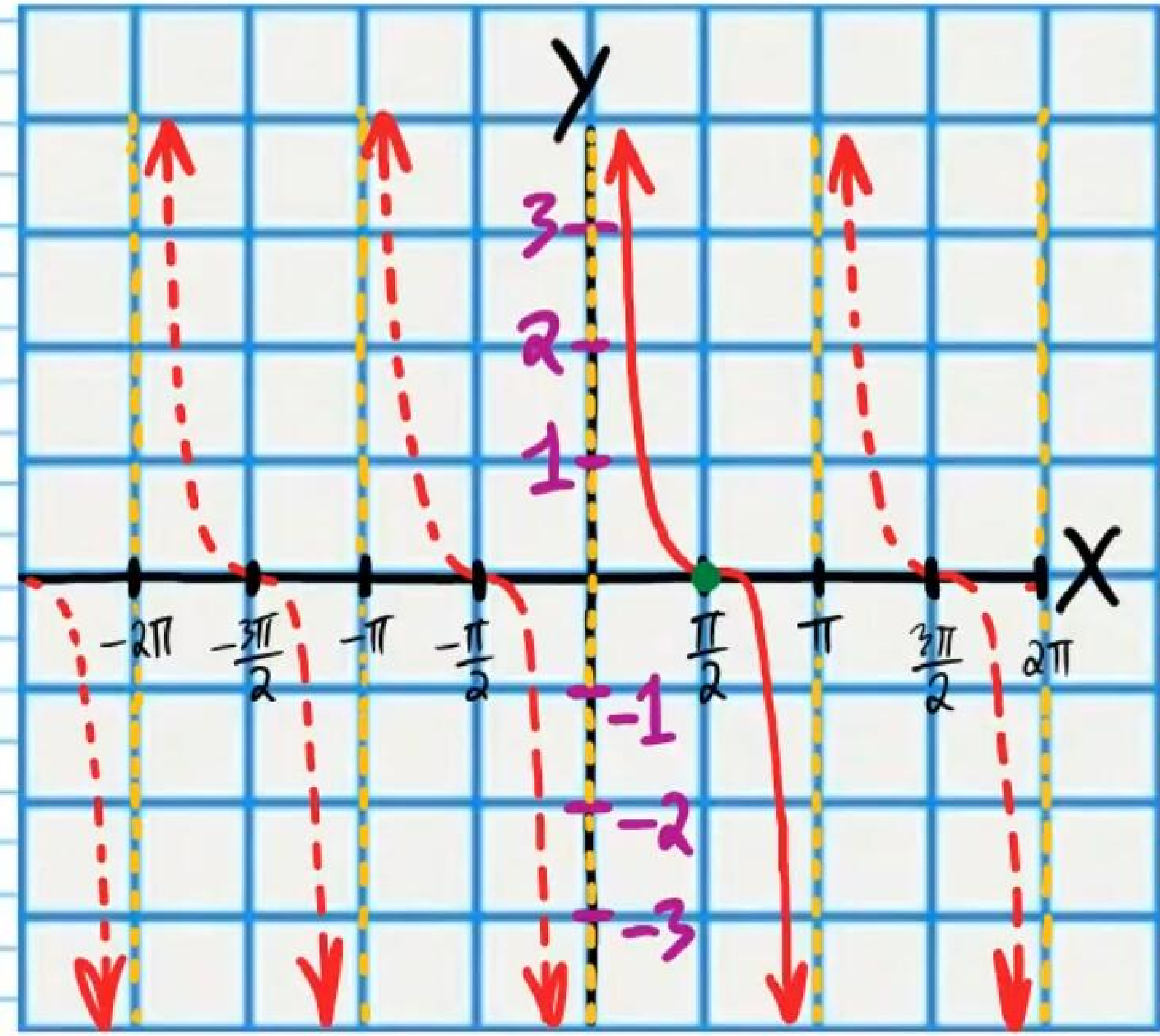
(82) $\tan 0 = \boxed{0}$

(83) $\tan(-\pi) = \boxed{0}$

(84) $\tan\left(-\frac{3\pi}{2}\right) = \boxed{\text{und}}$

(85) $\tan(2\pi) = \boxed{0}$

$y = \cot x$



Use the graph above to find the following values.

(86) $\cot\left(\frac{\pi}{2}\right) = \boxed{0}$

(87) $\cot \pi = \boxed{\text{und}}$

(88) $\cot\left(-\frac{\pi}{2}\right) = \boxed{0}$

(89) $\cot 0 = \boxed{\text{und}}$

(90) $\cot\left(\frac{3\pi}{2}\right) = \boxed{0}$